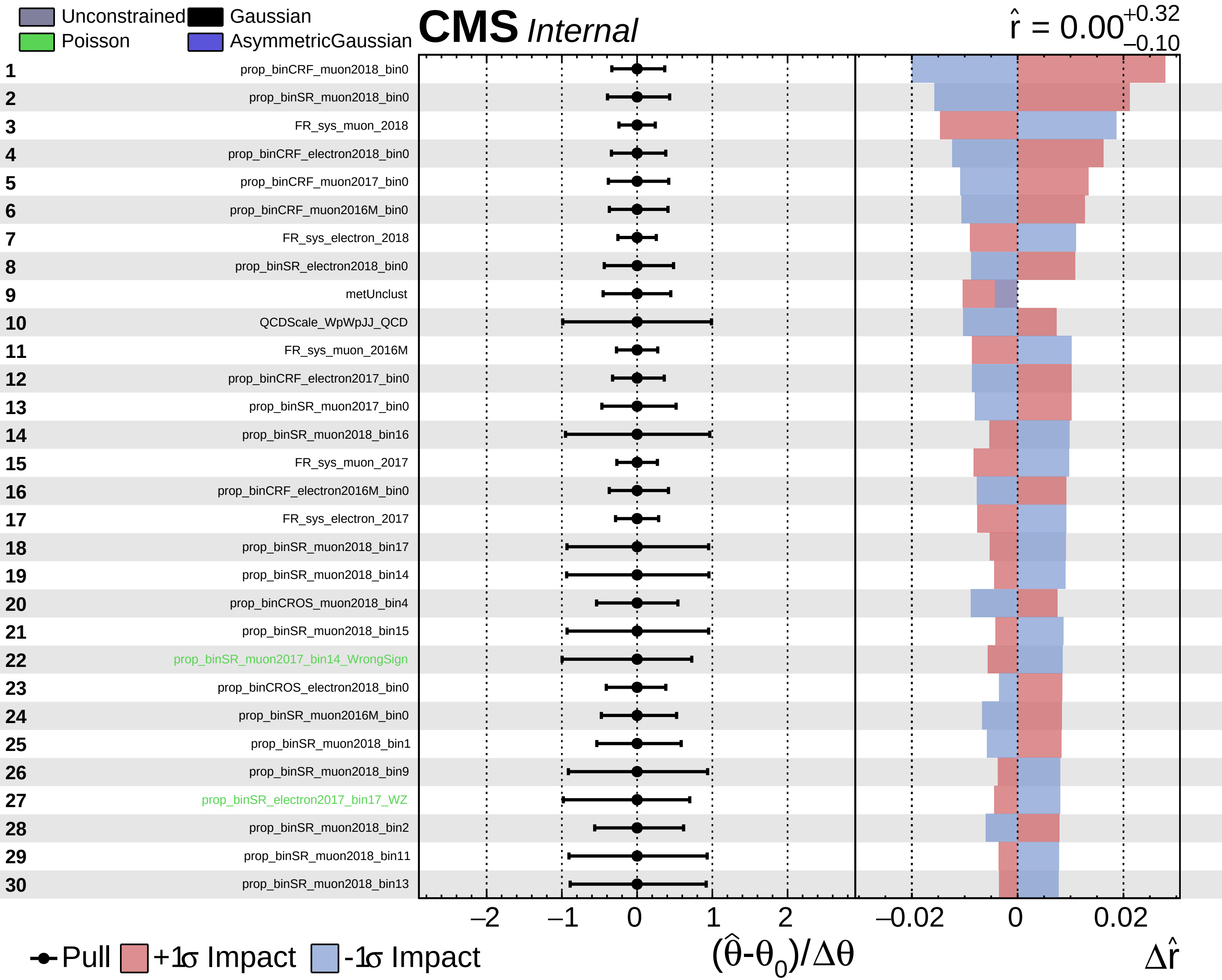
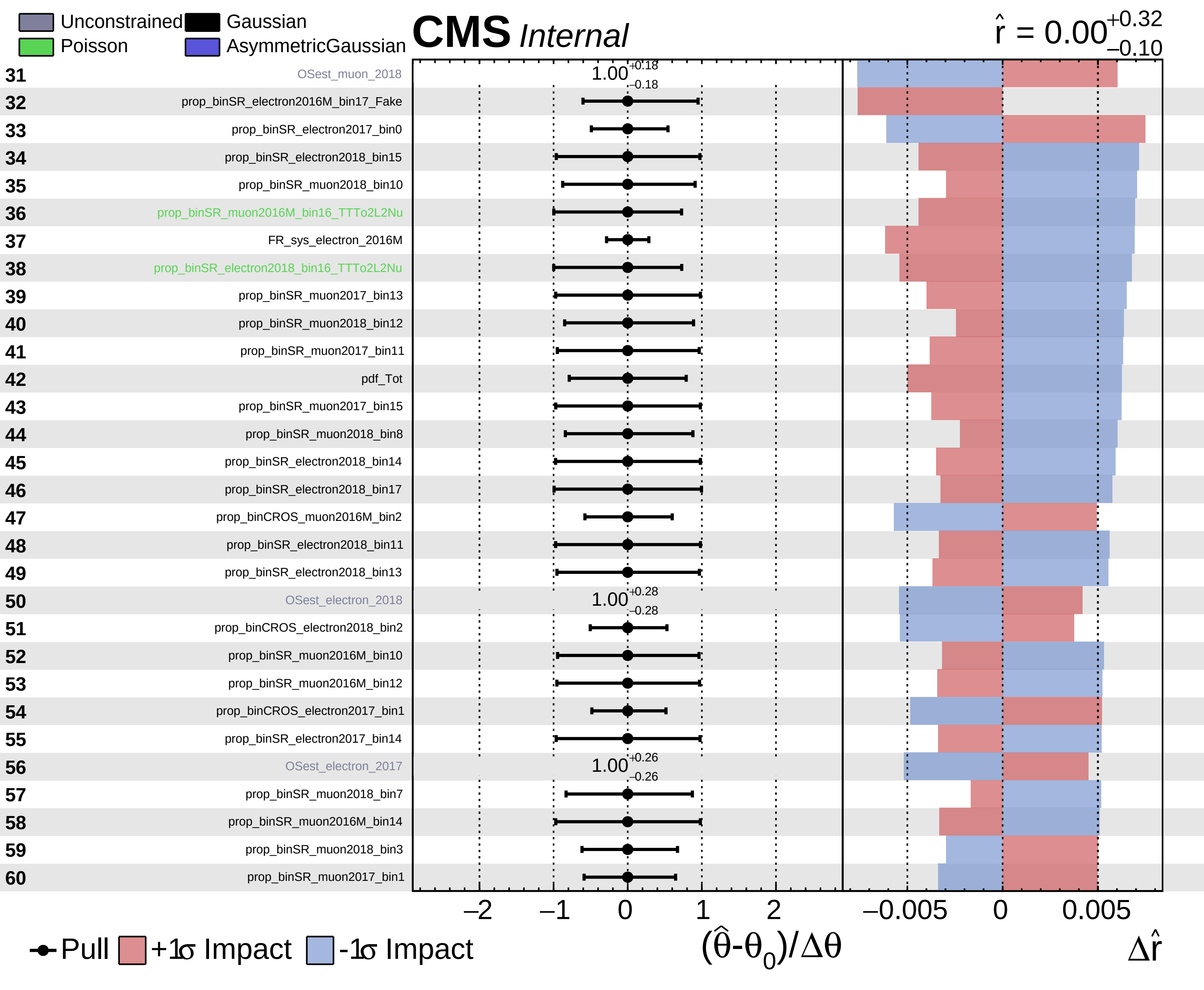
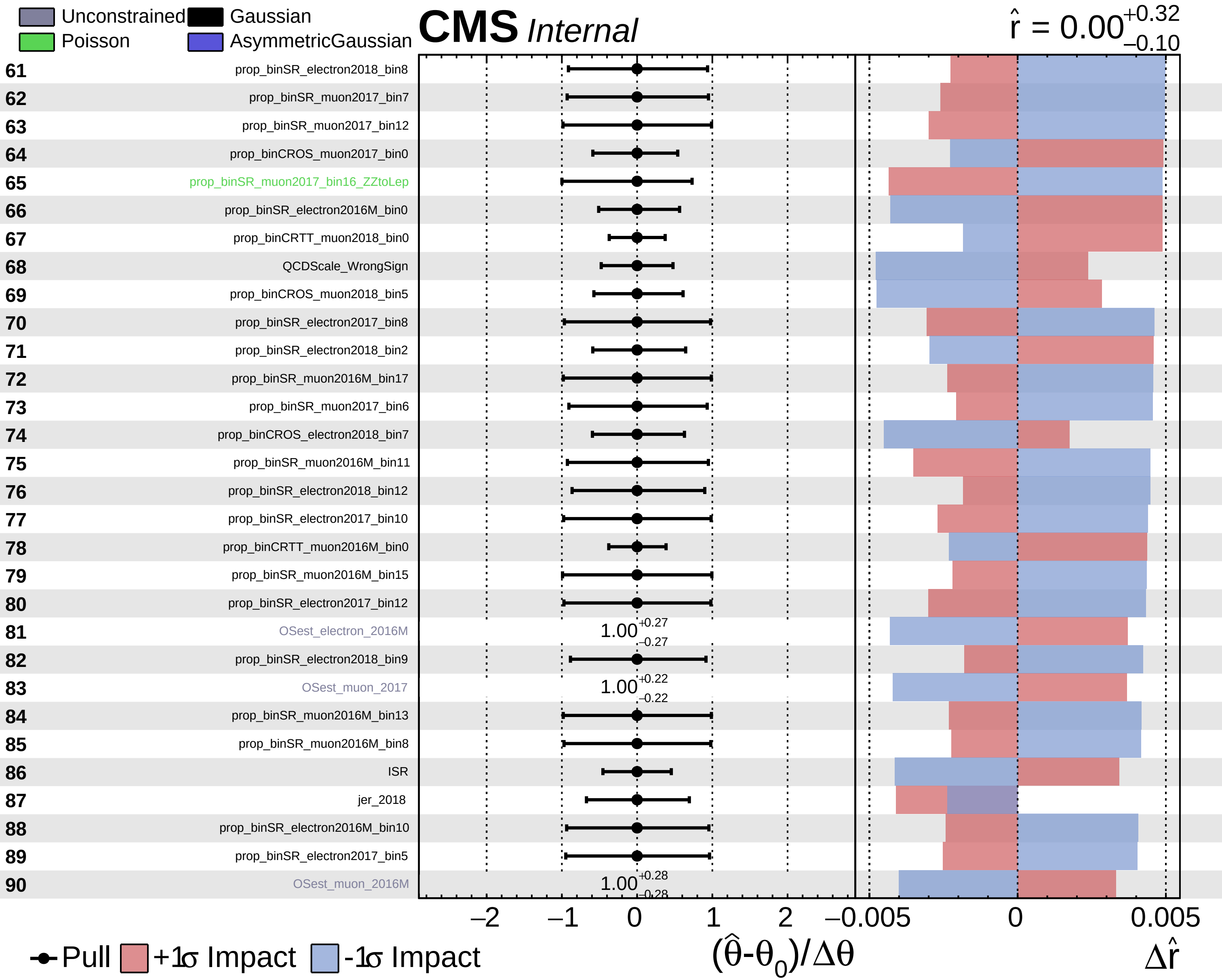
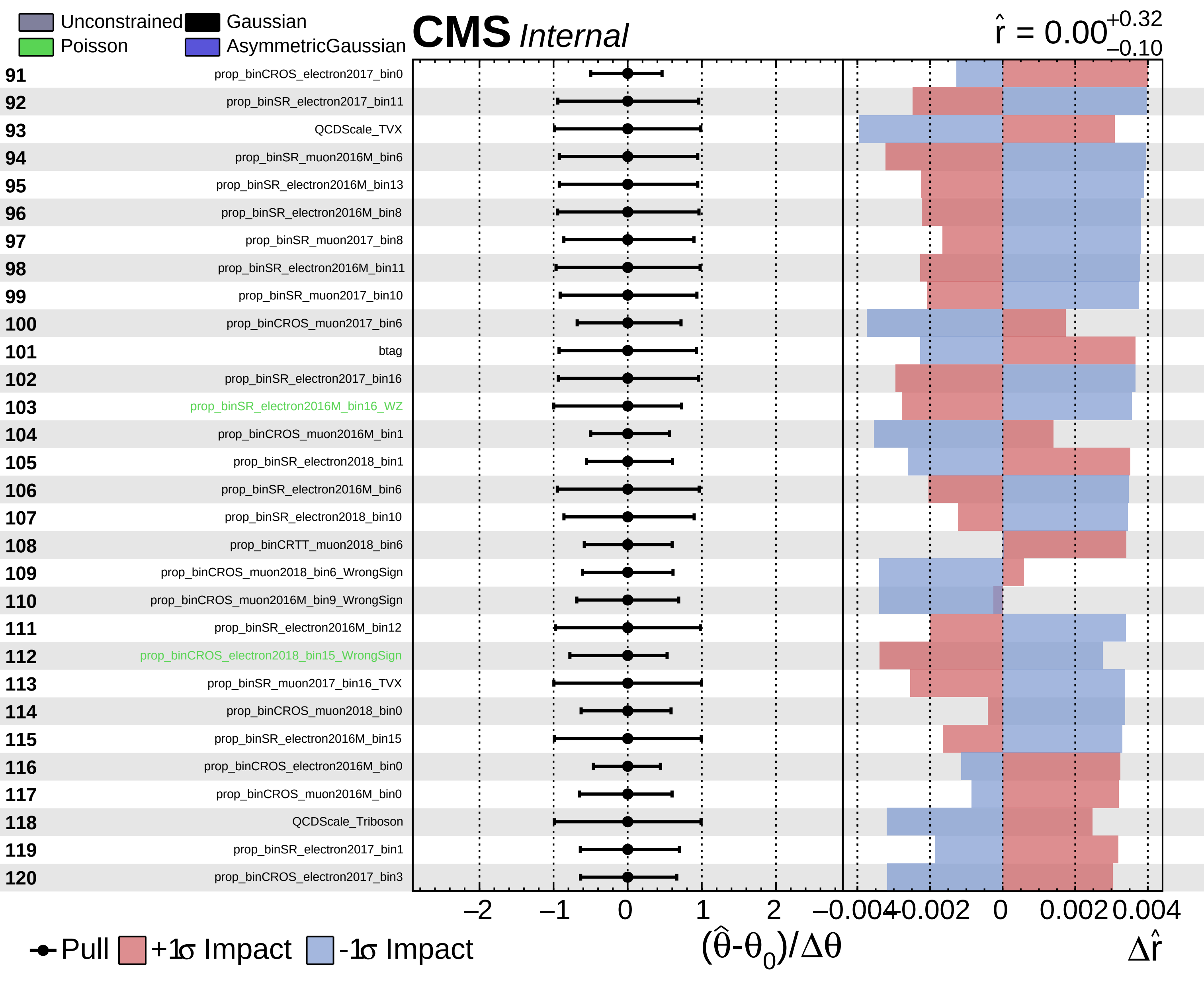

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
1 prop_binCRF_muon2018_bin0
2 prop_binSR_muon2018_bin0
3 FR_sys_muon_2018
4 prop_binCRF_electron2018_bin0
5 prop_binCRF_muon2017_bin0
6 prop_binCRF_muon2016M_bin0
7 FR_sys_electron_2018
8 prop_binSR_electron2018_bin0
9 metUnclust
10 QCDScale_WpWpJJ_QCD
11 FR_sys_muon_2016M
12 prop_binCRF_electron2017_bin0
13 prop_binSR_muon2017_bin0
14 prop_binSR_muon2018_bin16
15 FR_sys_muon_2017
16 prop_binCRF_electron2016M_bin0
17 FR_sys_electron_2017
18 prop_binSR_muon2018_bin17
19 prop_binSR_muon2018_bin14
20 prop_binCROS_muon2018_bin4
21 prop_binSR_muon2018_bin15
22 prop_binSR_muon2017_bin14_WrongSign
23 prop_binCROS_electron2018_bin0
24 prop_binSR_muon2016M_bin0
25 prop_binSR_muon2018_bin1
26 prop_binSR_muon2018_bin9
27 prop_binSR_electron2017_bin17_WZ
28 prop_binSR_muon2018_bin2
29 prop_binSR_muon2018_bin11
30 prop_binSR_muon2018_bin13
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

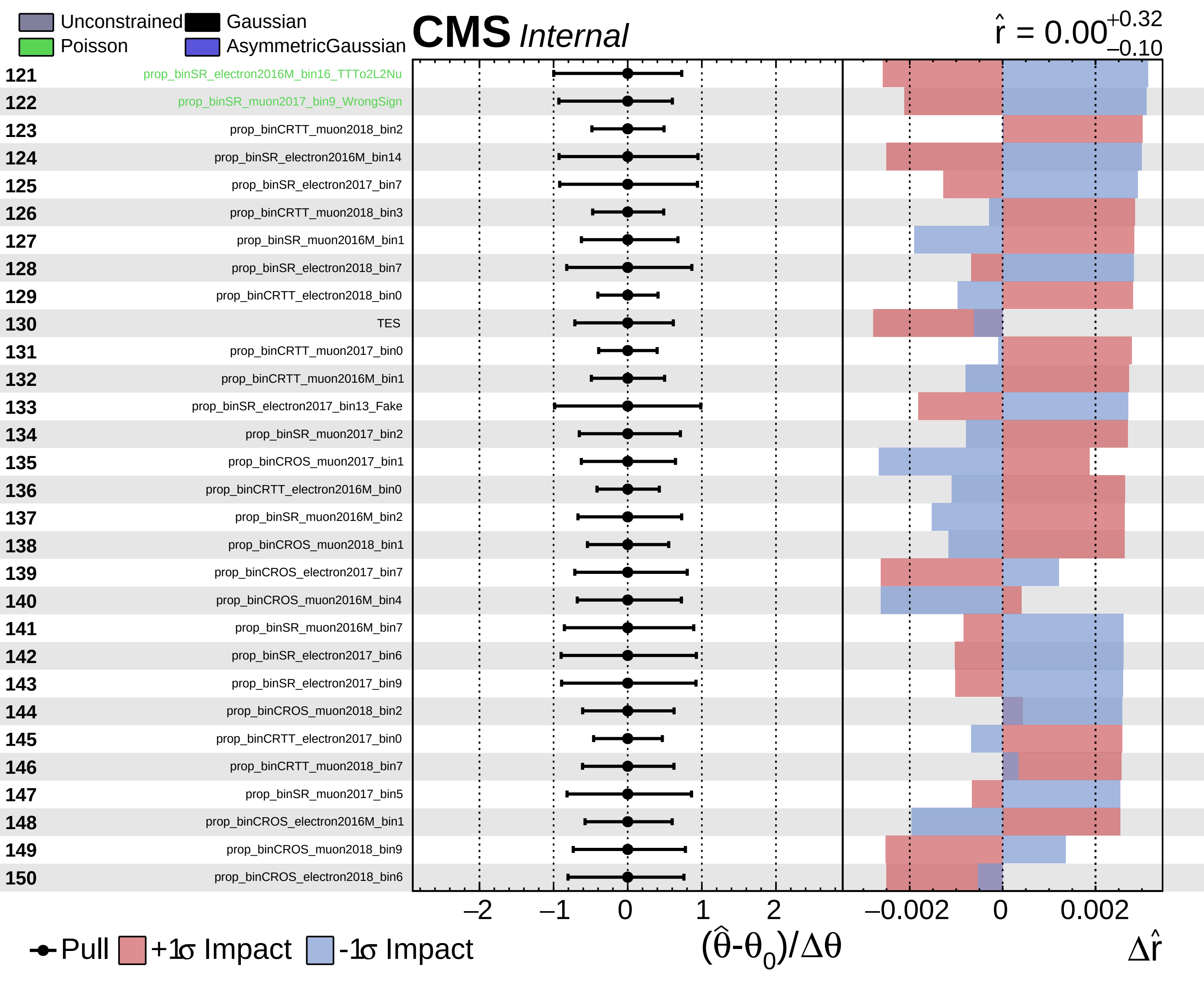


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.10
31 OSest_muon_2018 1.00 +0.18 −0.18
32 prop_binSR_electron2016M_bin17_Fake
33 prop_binSR_electron2017_bin0
34 prop_binSR_electron2018_bin15
35 prop_binSR_muon2018_bin10
36 prop_binSR_muon2016M_bin16_TTTo2L2Nu
37 FR_sys_electron_2016M
38 prop_binSR_electron2018_bin16_TTTo2L2Nu
39 prop_binSR_muon2017_bin13
40 prop_binSR_muon2018_bin12
41 prop_binSR_muon2017_bin11
42 pdf_Tot
43 prop_binSR_muon2017_bin15
44 prop_binSR_muon2018_bin8
45 prop_binSR_electron2018_bin14
46 prop_binSR_electron2018_bin17
47 prop_binCROS_muon2016M_bin2
48 prop_binSR_electron2018_bin11
49 prop_binSR_electron2018_bin13
50 OSest_electron_2018 1.00 +0.28 −0.28
51 prop_binCROS_electron2018_bin2
52 prop_binSR_muon2016M_bin10
53 prop_binSR_muon2016M_bin12
54 prop_binCROS_electron2017_bin1
55 prop_binSR_electron2017_bin14
56 OSest_electron_2017 1.00 +0.26 −0.26
57 prop_binSR_muon2018_bin7
58 prop_binSR_muon2016M_bin14
59 prop_binSR_muon2018_bin3
60 prop_binSR_muon2017_bin1
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

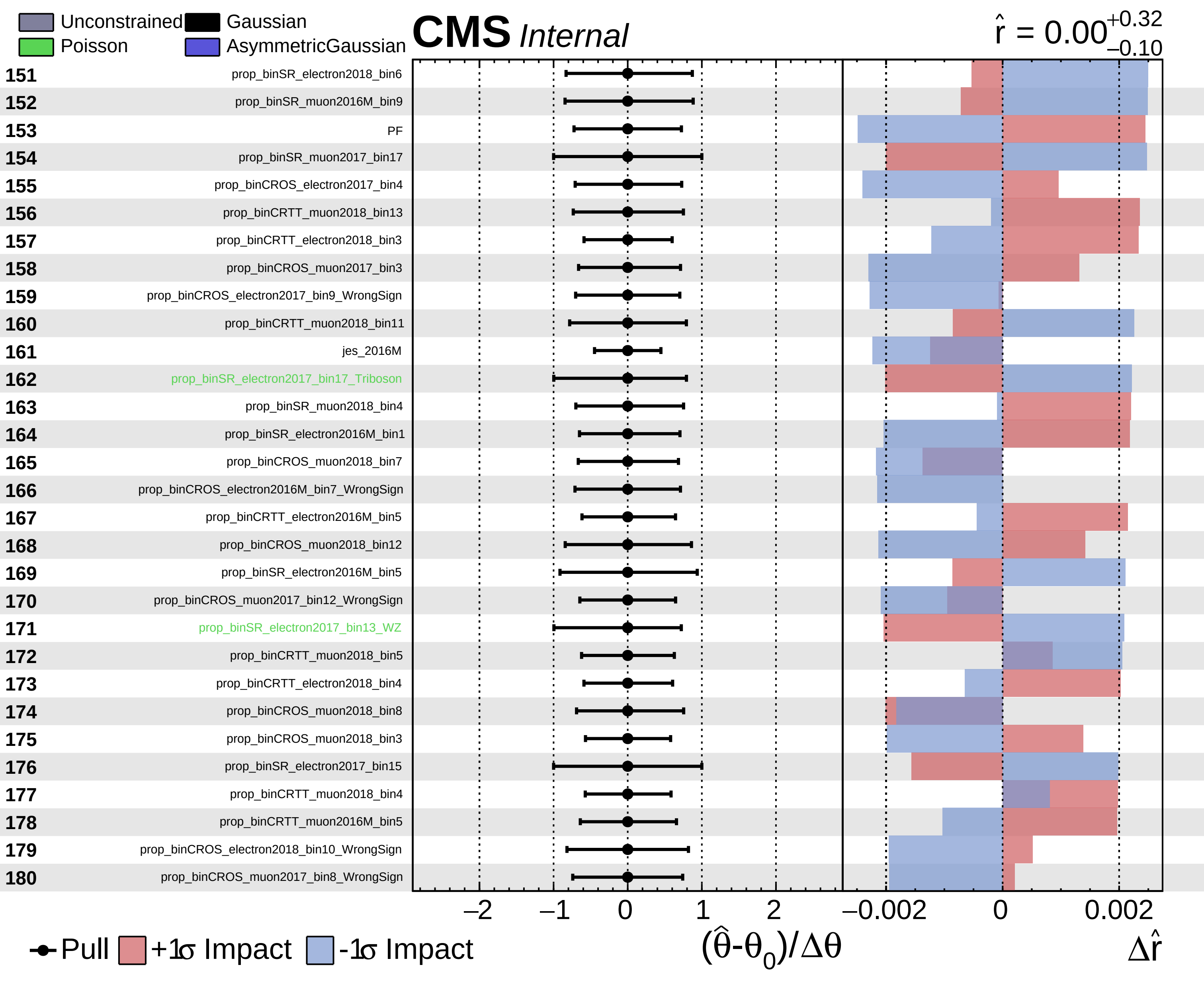


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
61 prop_binSR_electron2018_bin8
62 prop_binSR_muon2017_bin7
63 prop_binSR_muon2017_bin12
64 prop_binCROS_muon2017_bin0
65 prop_binSR_muon2017_bin16_ZZtoLep
66 prop_binSR_electron2016M_bin0
67 prop_binCRTT_muon2018_bin0
68 QCDScale_WrongSign
69 prop_binCROS_muon2018_bin5
70 prop_binSR_electron2017_bin8
71 prop_binSR_electron2018_bin2
72 prop_binSR_muon2016M_bin17
73 prop_binSR_muon2017_bin6
74 prop_binCROS_electron2018_bin7
75 prop_binSR_muon2016M_bin11
76 prop_binSR_electron2018_bin12
77 prop_binSR_electron2017_bin10
78 prop_binCRTT_muon2016M_bin0
79 prop_binSR_muon2016M_bin15
80 prop_binSR_electron2017_bin12
81 OSest_electron_2016M
$1.00^{+0.27}_{-0.27}$
82 prop_binSR_electron2018_bin9
83 OSest_muon_2017
$1.00^{+0.22}_{-0.22}$
84 prop_binSR_muon2016M_bin13
85 prop_binSR_muon2016M_bin8
86 ISR
87 jer_2018
88 prop_binSR_electron2016M_bin10
89 prop_binSR_electron2017_bin5
90 OSest_muon_2016M
$1.00^{+0.28}_{-0.28}$
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

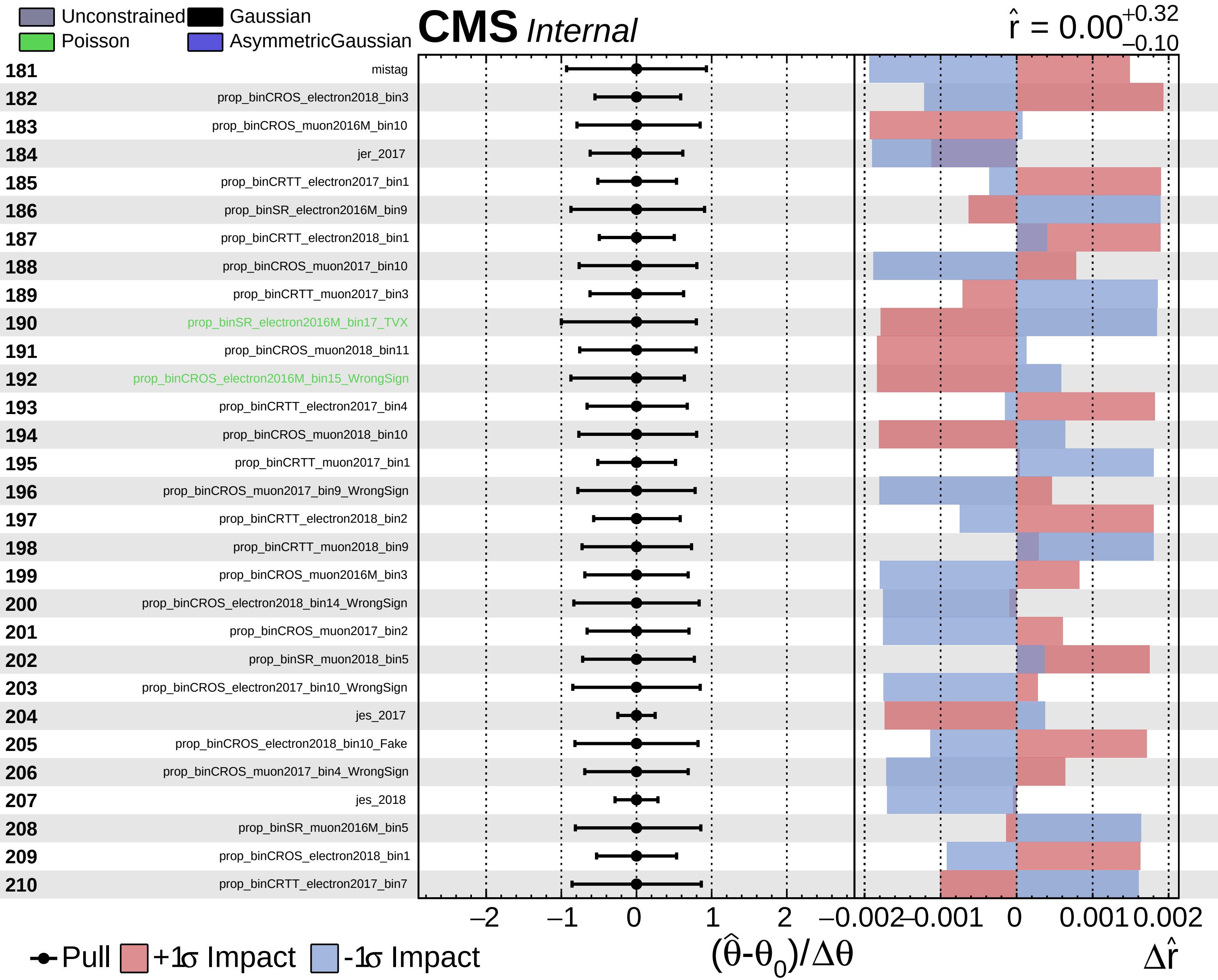

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
91 prop_binCROS_electron2017_bin0
92 prop_binSR_electron2017_bin11
93 QCDScale_TVX
94 prop_binSR_muon2016M_bin6
95 prop_binSR_electron2016M_bin13
96 prop_binSR_electron2016M_bin8
97 prop_binSR_muon2017_bin8
98 prop_binSR_electron2016M_bin11
99 prop_binSR_muon2017_bin10
100 prop_binCROS_muon2017_bin6
101 btag
102 prop_binSR_electron2017_bin16
103 prop_binSR_electron2016M_bin16_WZ
104 prop_binCROS_muon2016M_bin1
105 prop_binSR_electron2018_bin1
106 prop_binSR_electron2016M_bin6
107 prop_binSR_electron2018_bin10
108 prop_binCRTT_muon2018_bin6
109 prop_binCROS_muon2018_bin6_WrongSign
110 prop_binCROS_muon2016M_bin9_WrongSign
111 prop_binSR_electron2016M_bin12
112 prop_binCROS_electron2018_bin15_WrongSign
113 prop_binSR_muon2017_bin16_TVX
114 prop_binCROS_muon2018_bin0
115 prop_binSR_electron2016M_bin15
116 prop_binCROS_electron2016M_bin0
117 prop_binCROS_muon2016M_bin0
118 QCDScale_Triboson
119 prop_binSR_electron2017_bin1
120 prop_binCROS_electron2017_bin3
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

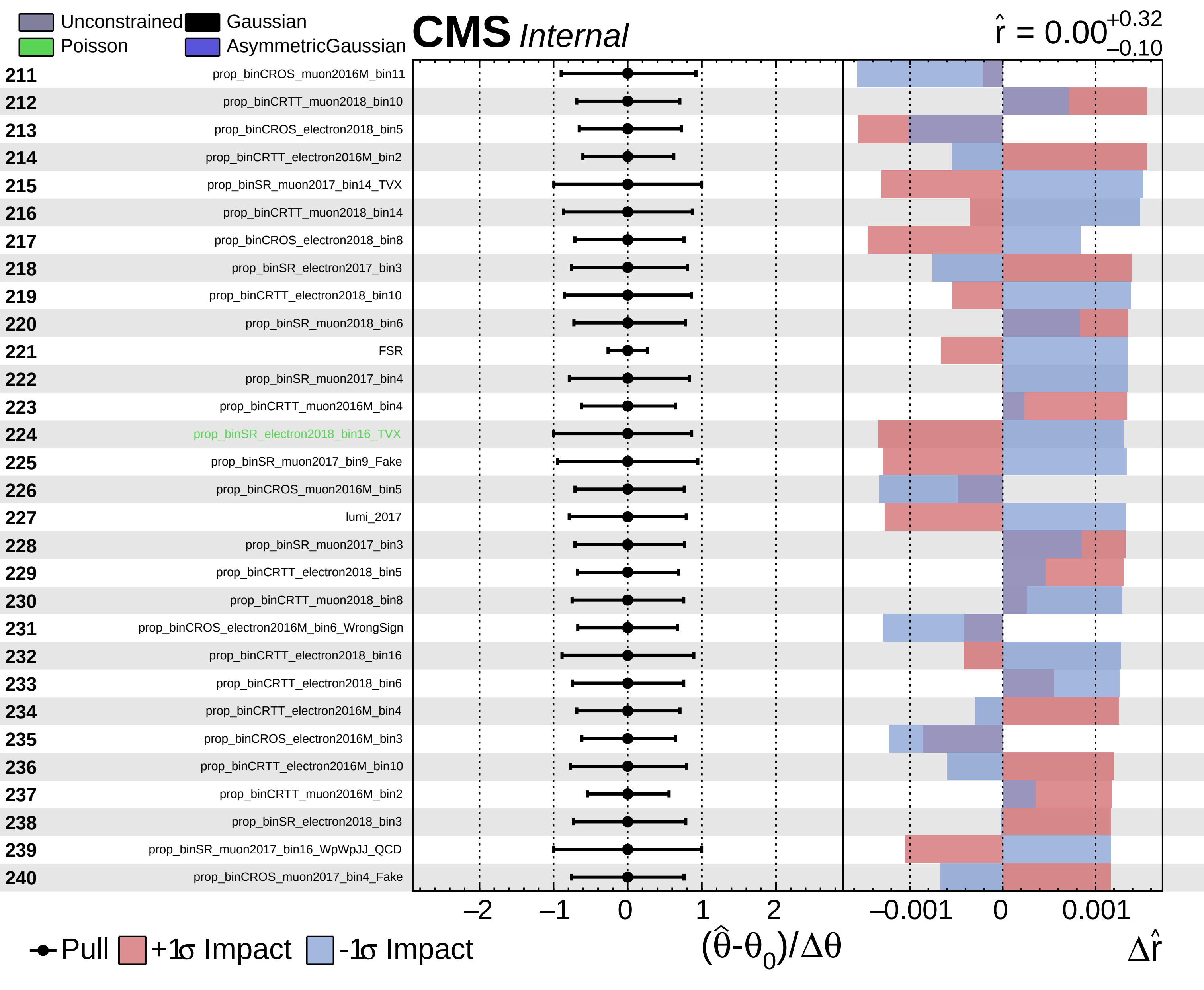


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
121 prop_binSR_electron2016M_bin16_TTTo2L2Nu
122 prop_binSR_muon2017_bin9_WrongSign
123 prop_binCRTT_muon2018_bin2
124 prop_binSR_electron2016M_bin14
125 prop_binSR_electron2017_bin7
126 prop_binCRTT_muon2018_bin3
127 prop_binSR_muon2016M_bin1
128 prop_binSR_electron2018_bin7
129 prop_binCRTT_electron2018_bin0
130 TES
131 prop_binCRTT_muon2017_bin0
132 prop_binCRTT_muon2016M_bin1
133 prop_binSR_electron2017_bin13_Fake
134 prop_binSR_muon2017_bin2
135 prop_binCROS_muon2017_bin1
136 prop_binCRTT_electron2016M_bin0
137 prop_binSR_muon2016M_bin2
138 prop_binCROS_muon2018_bin1
139 prop_binCROS_electron2017_bin7
140 prop_binCROS_muon2016M_bin4
141 prop_binSR_muon2016M_bin7
142 prop_binSR_electron2017_bin6
143 prop_binSR_electron2017_bin9
144 prop_binCROS_muon2018_bin2
145 prop_binCRTT_electron2017_bin0
146 prop_binCRTT_muon2018_bin7
147 prop_binSR_muon2017_bin5
148 prop_binCROS_electron2016M_bin1
149 prop_binCROS_muon2018_bin9
150 prop_binCROS_electron2018_bin6
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

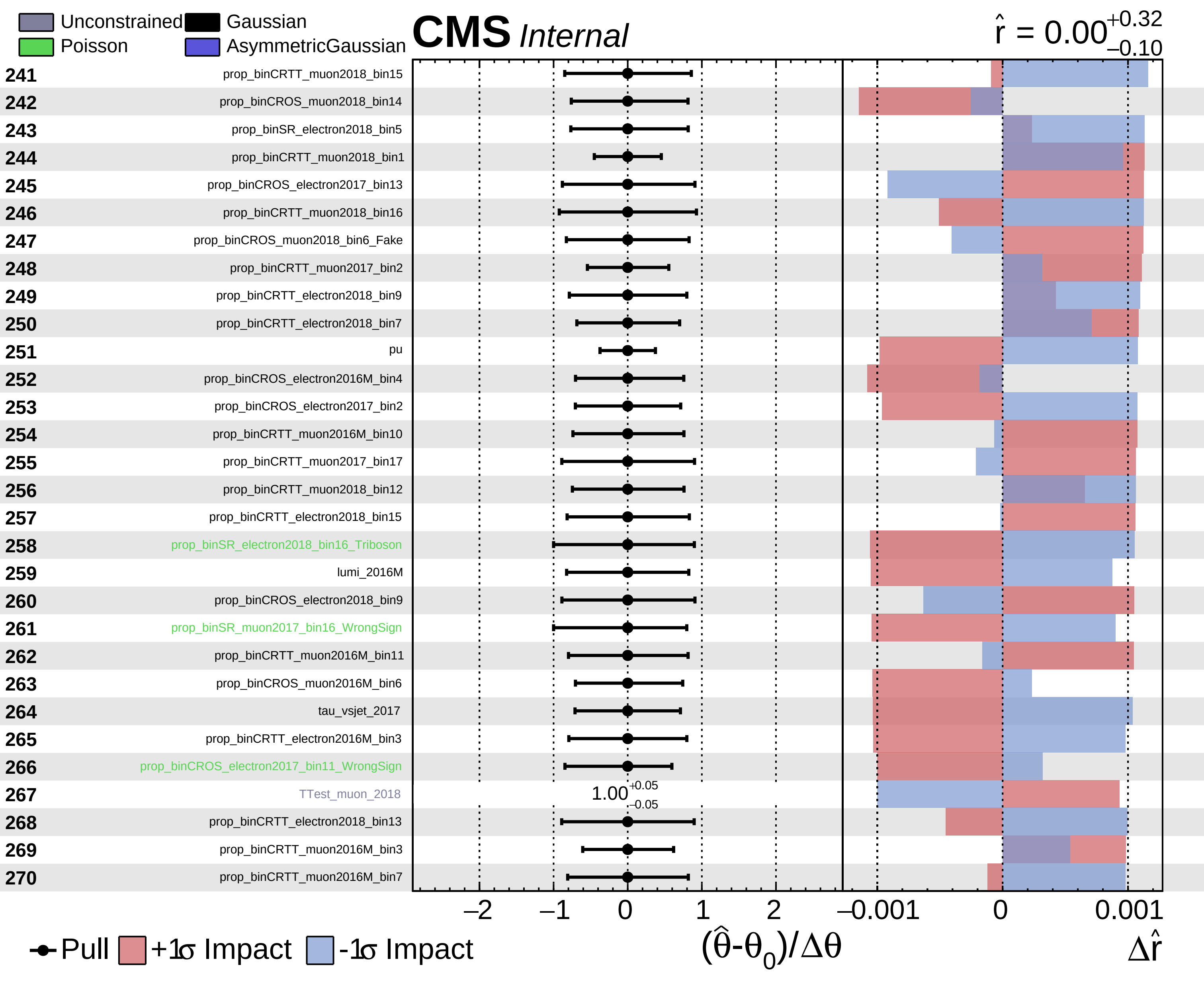


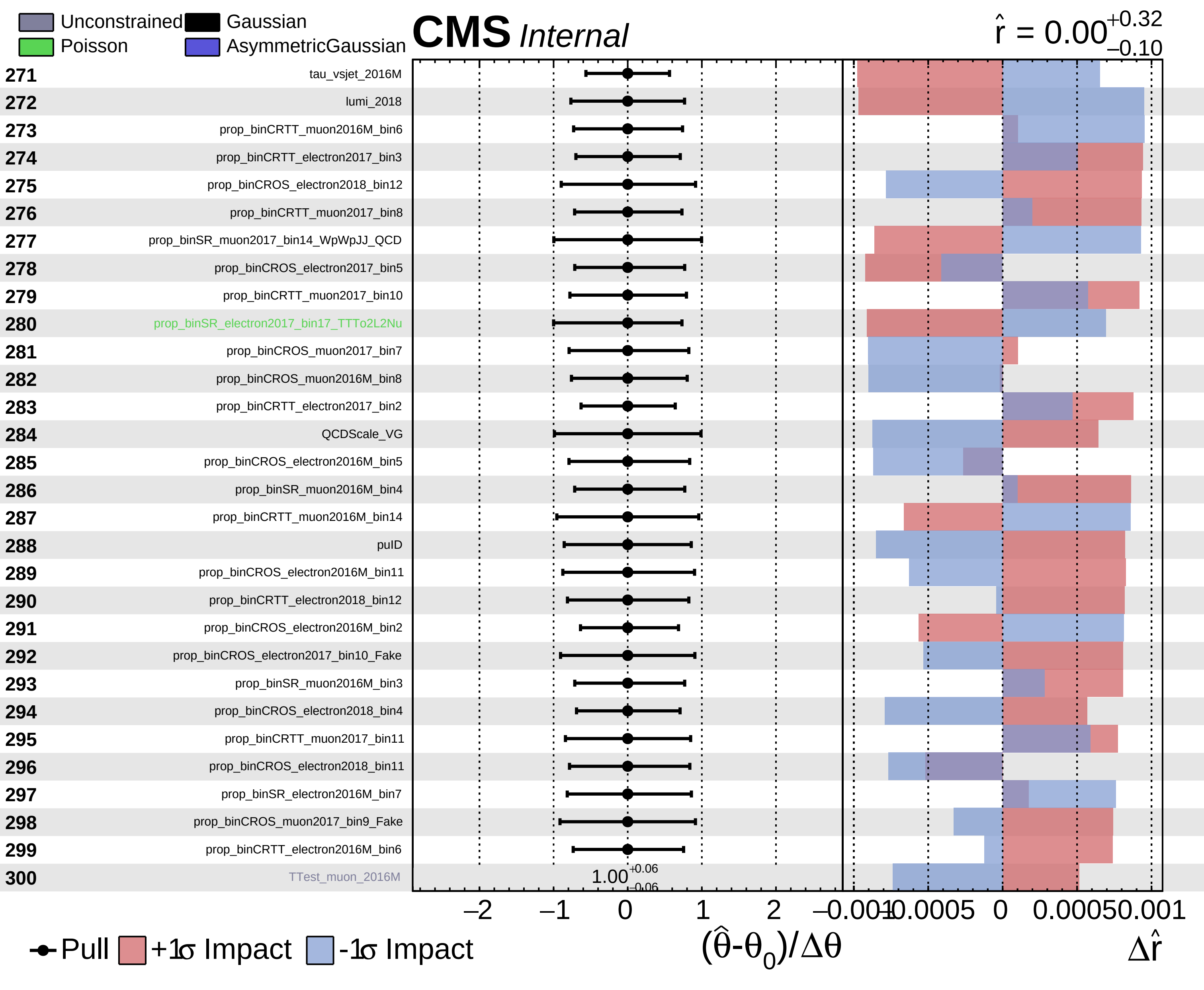
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
151 prop_binSR_electron2018_bin6
152 prop_binSR_muon2016M_bin9
153 PF
154 prop_binSR_muon2017_bin17
155 prop_binCROS_electron2017_bin4
156 prop_binCRTT_muon2018_bin13
157 prop_binCRTT_electron2018_bin3
158 prop_binCROS_muon2017_bin3
159 prop_binCROS_electron2017_bin9_WrongSign
160 prop_binCRTT_muon2018_bin11
161 jes_2016M
162 prop_binSR_electron2017_bin17_Triboson
163 prop_binSR_muon2018_bin4
164 prop_binSR_electron2016M_bin1
165 prop_binCROS_muon2018_bin7
166 prop_binCROS_electron2016M_bin7_WrongSign
167 prop_binCRTT_electron2016M_bin5
168 prop_binCROS_muon2018_bin12
169 prop_binSR_electron2016M_bin5
170 prop_binCROS_muon2017_bin12_WrongSign
171 prop_binSR_electron2017_bin13_WZ
172 prop_binCRTT_muon2018_bin5
173 prop_binCRTT_electron2018_bin4
174 prop_binCROS_muon2018_bin8
175 prop_binCROS_muon2018_bin3
176 prop_binSR_electron2017_bin15
177 prop_binCRTT_muon2018_bin4
178 prop_binCRTT_muon2016M_bin5
179 prop_binCROS_electron2018_bin10_WrongSign
180 prop_binCROS_muon2017_bin8_WrongSign
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
181 mistag
182 prop_binCROS_electron2018_bin3
183 prop_binCROS_muon2016M_bin10
184 jer_2017
185 prop_binCRTT_electron2017_bin1
186 prop_binSR_electron2016M_bin9
187 prop_binCRTT_electron2018_bin1
188 prop_binCROS_muon2017_bin10
189 prop_binCRTT_muon2017_bin3
190 prop_binSR_electron2016M_bin17_TVX
191 prop_binCROS_muon2018_bin11
192 prop_binCROS_electron2016M_bin15_WrongSign
193 prop_binCRTT_electron2017_bin4
194 prop_binCROS_muon2018_bin10
195 prop_binCRTT_muon2017_bin1
196 prop_binCROS_muon2017_bin9_WrongSign
197 prop_binCRTT_electron2018_bin2
198 prop_binCRTT_muon2018_bin9
199 prop_binCROS_muon2016M_bin3
200 prop_binCROS_electron2018_bin14_WrongSign
201 prop_binCROS_muon2017_bin2
202 prop_binSR_muon2018_bin5
203 prop_binCROS_electron2017_bin10_WrongSign
204 jes_2017
205 prop_binCROS_electron2018_bin10_Fake
206 prop_binCROS_muon2017_bin4_WrongSign
207 jes_2018
208 prop_binSR_muon2016M_bin5
209 prop_binCROS_electron2018_bin1
210 prop_binCRTT_electron2017_bin7
−2
−1
0
1
2
−0.002
−0.001
0
0.001
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

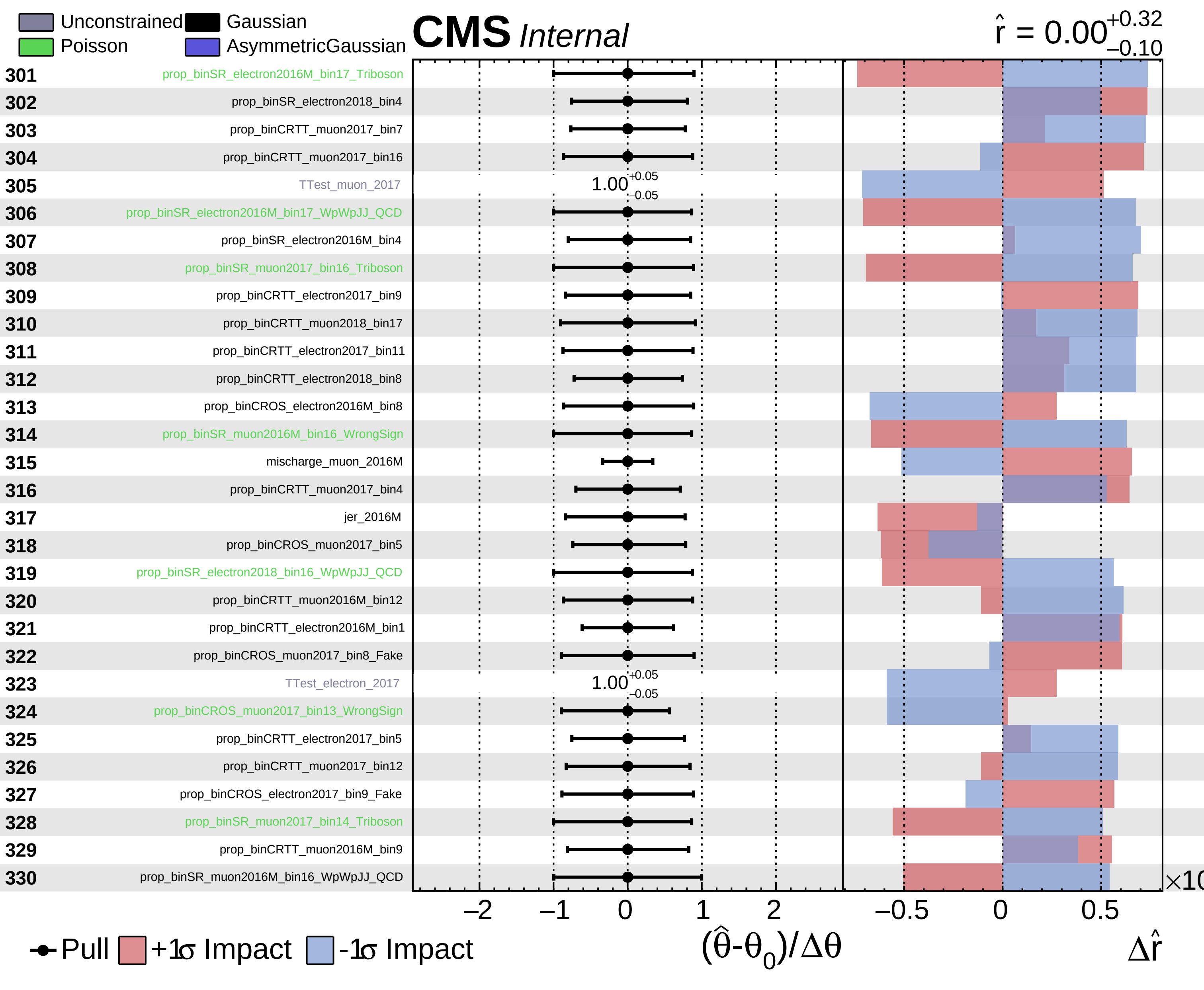


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
211 prop_binCROS_muon2016M_bin11
212 prop_binCRTT_muon2018_bin10
213 prop_binCROS_electron2018_bin5
214 prop_binCRTT_electron2016M_bin2
215 prop_binSR_muon2017_bin14_TVX
216 prop_binCRTT_muon2018_bin14
217 prop_binCROS_electron2018_bin8
218 prop_binSR_electron2017_bin3
219 prop_binCRTT_electron2018_bin10
220 prop_binSR_muon2018_bin6
221 FSR
222 prop_binSR_muon2017_bin4
223 prop_binCRTT_muon2016M_bin4
224 prop_binSR_electron2018_bin16_TVX
225 prop_binSR_muon2017_bin9_Fake
226 prop_binCROS_muon2016M_bin5
227 lumi_2017
228 prop_binSR_muon2017_bin3
229 prop_binCRTT_electron2018_bin5
230 prop_binCRTT_muon2018_bin8
231 prop_binCROS_electron2016M_bin6_WrongSign
232 prop_binCRTT_electron2018_bin16
233 prop_binCRTT_electron2018_bin6
234 prop_binCRTT_electron2016M_bin4
235 prop_binCROS_electron2016M_bin3
236 prop_binCRTT_electron2016M_bin10
237 prop_binCRTT_muon2016M_bin2
238 prop_binSR_electron2018_bin3
239 prop_binSR_muon2017_bin16_WpWpJJ_QCD
240 prop_binCROS_muon2017_bin4_Fake
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

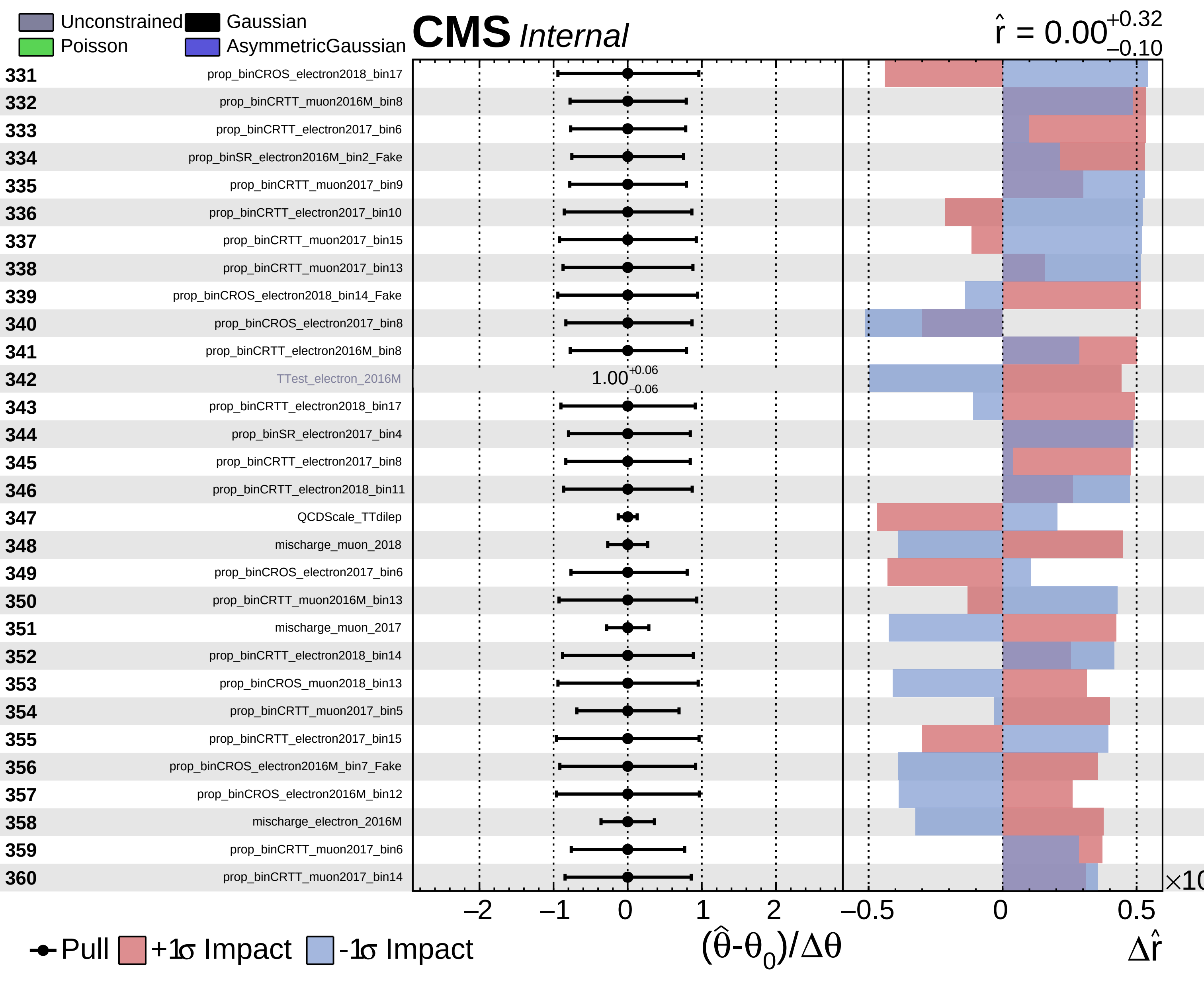

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
241 prop_binCRTT_muon2018_bin15
242 prop_binCROS_muon2018_bin14
243 prop_binSR_electron2018_bin5
244 prop_binCRTT_muon2018_bin1
245 prop_binCROS_electron2017_bin13
246 prop_binCRTT_muon2018_bin16
247 prop_binCROS_muon2018_bin6_Fake
248 prop_binCRTT_muon2017_bin2
249 prop_binCRTT_electron2018_bin9
250 prop_binCRTT_electron2018_bin7
251 pu
252 prop_binCROS_electron2016M_bin4
253 prop_binCROS_electron2017_bin2
254 prop_binCRTT_muon2016M_bin10
255 prop_binCRTT_muon2017_bin17
256 prop_binCRTT_muon2018_bin12
257 prop_binCRTT_electron2018_bin15
258 prop_binSR_electron2018_bin16_Triboson
259 lumi_2016M
260 prop_binCROS_electron2018_bin9
261 prop_binSR_muon2017_bin16_WrongSign
262 prop_binCRTT_muon2016M_bin11
263 prop_binCROS_muon2016M_bin6
264 tau_vsjet_2017
265 prop_binCRTT_electron2016M_bin3
266 prop_binCROS_electron2017_bin11_WrongSign
267 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
268 prop_binCRTT_electron2018_bin13
269 prop_binCRTT_muon2016M_bin3
270 prop_binCRTT_muon2016M_bin7
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

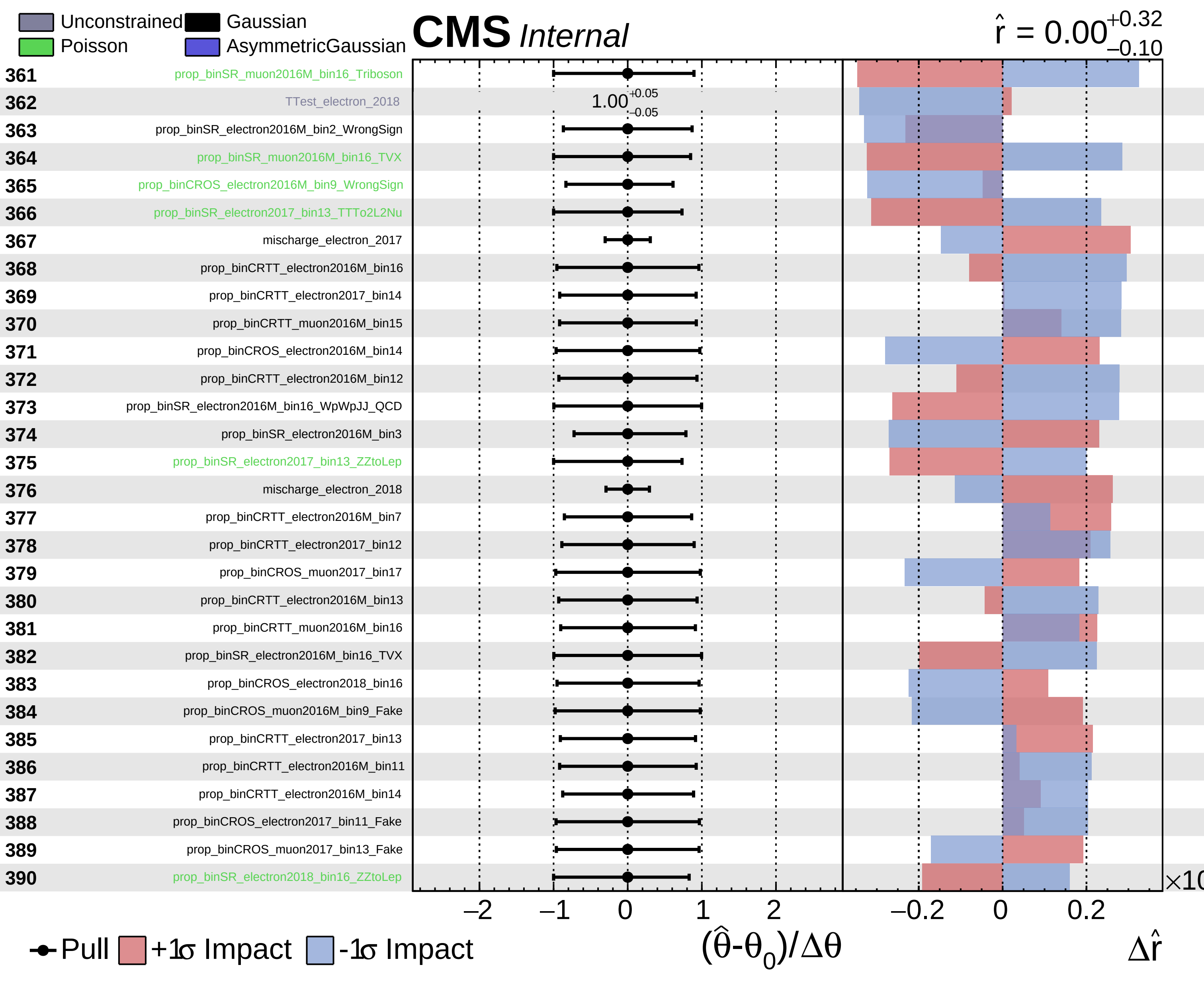


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.10
271 tau_vsjet_2016M
272 lumi_2018
273 prop_binCRTT_muon2016M_bin6
274 prop_binCRTT_electron2017_bin3
275 prop_binCROS_electron2018_bin12
276 prop_binCRTT_muon2017_bin8
277 prop_binSR_muon2017_bin14_WpWpJJ_QCD
278 prop_binCROS_electron2017_bin5
279 prop_binCRTT_muon2017_bin10
280 prop_binSR_electron2017_bin17_TTTo2L2Nu
281 prop_binCROS_muon2017_bin7
282 prop_binCROS_muon2016M_bin8
283 prop_binCRTT_electron2017_bin2
284 QCDScale_VG
285 prop_binCROS_electron2016M_bin5
286 prop_binSR_muon2016M_bin4
287 prop_binCRTT_muon2016M_bin14
288 puID
289 prop_binCROS_electron2016M_bin11
290 prop_binCRTT_electron2018_bin12
291 prop_binCROS_electron2016M_bin2
292 prop_binCROS_electron2017_bin10_Fake
293 prop_binSR_muon2016M_bin3
294 prop_binCROS_electron2018_bin4
295 prop_binCRTT_muon2017_bin11
296 prop_binCROS_electron2018_bin11
297 prop_binSR_electron2016M_bin7
298 prop_binCROS_muon2017_bin9_Fake
299 prop_binCRTT_electron2016M_bin6
300 TTest_muon_2016M
1.00 +0.06 −0.06
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

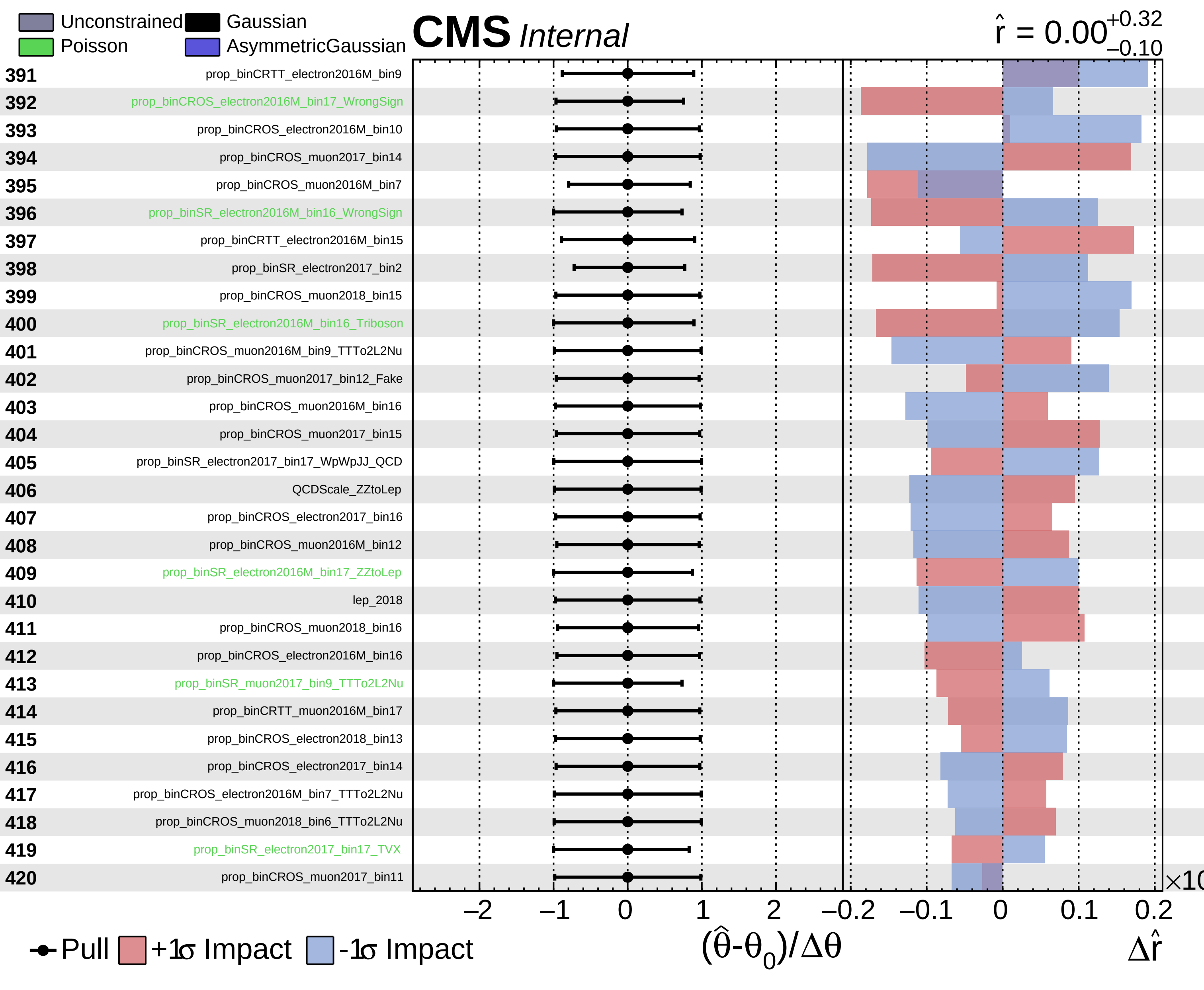


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.10
301 prop_binSR_electron2016M_bin17_Triboson
302 prop_binSR_electron2018_bin4
303 prop_binCRTT_muon2017_bin7
304 prop_binCRTT_muon2017_bin16
305 TTest_muon_2017
1.00 +0.05 −0.05
306 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
307 prop_binSR_electron2016M_bin4
308 prop_binSR_muon2017_bin16_Triboson
309 prop_binCRTT_electron2017_bin9
310 prop_binCRTT_muon2018_bin17
311 prop_binCRTT_electron2017_bin11
312 prop_binCRTT_electron2018_bin8
313 prop_binCROS_electron2016M_bin8
314 prop_binSR_muon2016M_bin16_WrongSign
315 mischarge_muon_2016M
316 prop_binCRTT_muon2017_bin4
317 jer_2016M
318 prop_binCROS_muon2017_bin5
319 prop_binSR_electron2018_bin16_WpWpJJ_QCD
320 prop_binCRTT_muon2016M_bin12
321 prop_binCRTT_electron2016M_bin1
322 prop_binCROS_muon2017_bin8_Fake
323 TTest_electron_2017
1.00 +0.05 −0.05
324 prop_binCROS_muon2017_bin13_WrongSign
325 prop_binCRTT_electron2017_bin5
326 prop_binCRTT_muon2017_bin12
327 prop_binCROS_electron2017_bin9_Fake
328 prop_binSR_muon2017_bin14_Triboson
329 prop_binCRTT_muon2016M_bin9
330 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

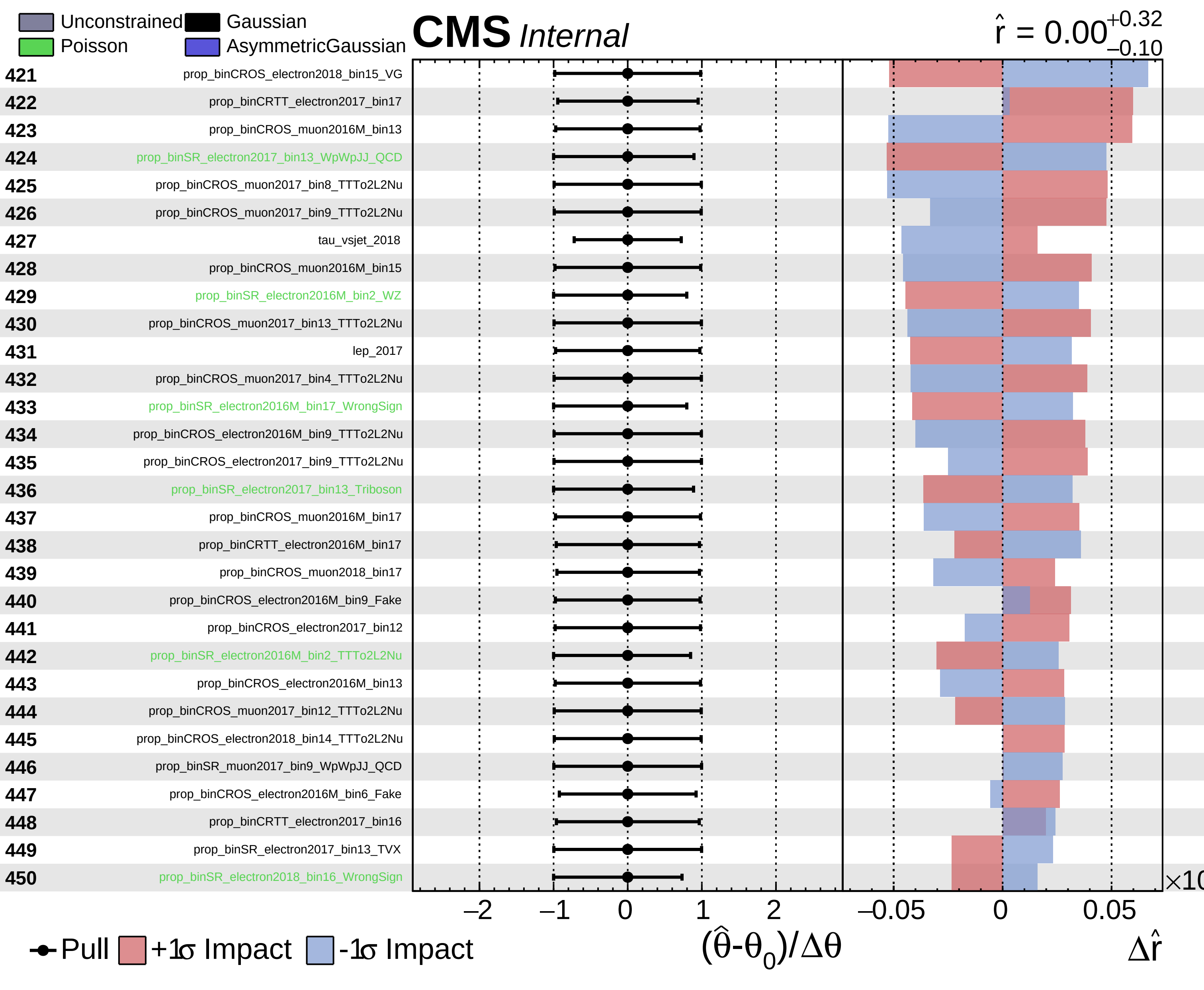

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
331 prop_binCROS_electron2018_bin17
332 prop_binCRTT_muon2016M_bin8
333 prop_binCRTT_electron2017_bin6
334 prop_binSR_electron2016M_bin2_Fake
335 prop_binCRTT_muon2017_bin9
336 prop_binCRTT_electron2017_bin10
337 prop_binCRTT_muon2017_bin15
338 prop_binCRTT_muon2017_bin13
339 prop_binCROS_electron2018_bin14_Fake
340 prop_binCROS_electron2017_bin8
341 prop_binCRTT_electron2016M_bin8
342 TTest_electron_2016M
$1.00^{+0.06}_{-0.06}$
343 prop_binCRTT_electron2018_bin17
344 prop_binSR_electron2017_bin4
345 prop_binCRTT_electron2017_bin8
346 prop_binCRTT_electron2018_bin11
347 QCDScale_TTdilep
348 mischarge_muon_2018
349 prop_binCROS_electron2017_bin6
350 prop_binCRTT_muon2016M_bin13
351 mischarge_muon_2017
352 prop_binCRTT_electron2018_bin14
353 prop_binCROS_muon2018_bin13
354 prop_binCRTT_muon2017_bin5
355 prop_binCRTT_electron2017_bin15
356 prop_binCROS_electron2016M_bin7_Fake
357 prop_binCROS_electron2016M_bin12
358 mischarge_electron_2016M
359 prop_binCRTT_muon2017_bin6
360 prop_binCRTT_muon2017_bin14
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



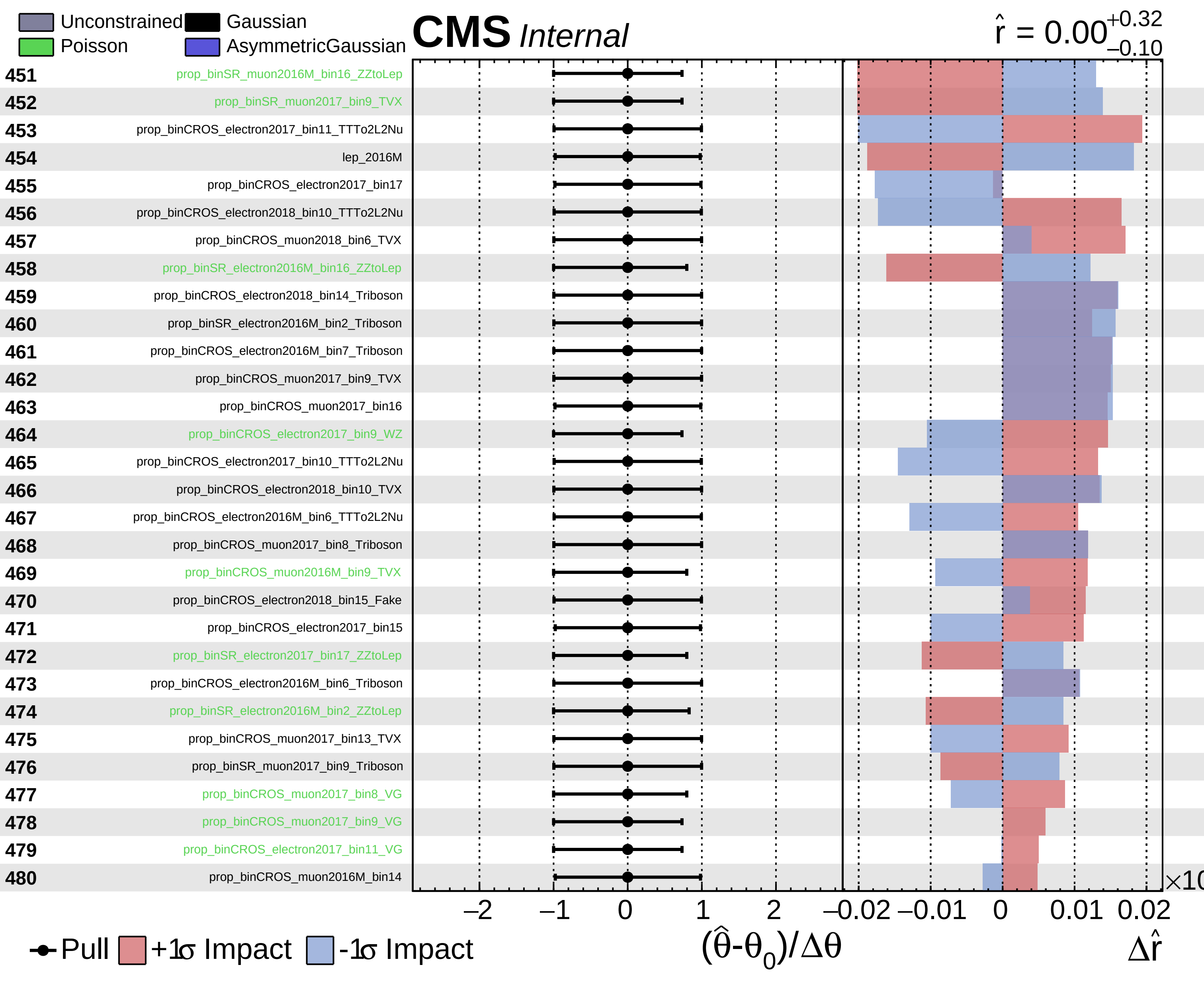
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
361 prop_binSR_muon2016M_bin16_Triboson
362 TTest_electron_2018
1.00 +0.05 −0.05
363 prop_binSR_electron2016M_bin2_WrongSign
364 prop_binSR_muon2016M_bin16_TVX
365 prop_binCROS_electron2016M_bin9_WrongSign
366 prop_binSR_electron2017_bin13_TTTo2L2Nu
367 mischarge_electron_2017
368 prop_binCRTT_electron2016M_bin16
369 prop_binCRTT_electron2017_bin14
370 prop_binCRTT_muon2016M_bin15
371 prop_binCROS_electron2016M_bin14
372 prop_binCRTT_electron2016M_bin12
373 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
374 prop_binSR_electron2016M_bin3
375 prop_binSR_electron2017_bin13_ZZtoLep
376 mischarge_electron_2018
377 prop_binCRTT_electron2016M_bin7
378 prop_binCRTT_electron2017_bin12
379 prop_binCROS_muon2017_bin17
380 prop_binCRTT_electron2016M_bin13
381 prop_binCRTT_muon2016M_bin16
382 prop_binSR_electron2016M_bin16_TVX
383 prop_binCROS_electron2018_bin16
384 prop_binCROS_muon2016M_bin9_Fake
385 prop_binCRTT_electron2017_bin13
386 prop_binCRTT_electron2016M_bin11
387 prop_binCRTT_electron2016M_bin14
388 prop_binCROS_electron2017_bin11_Fake
389 prop_binCROS_muon2017_bin13_Fake
390 prop_binSR_electron2018_bin16_ZZtoLep
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



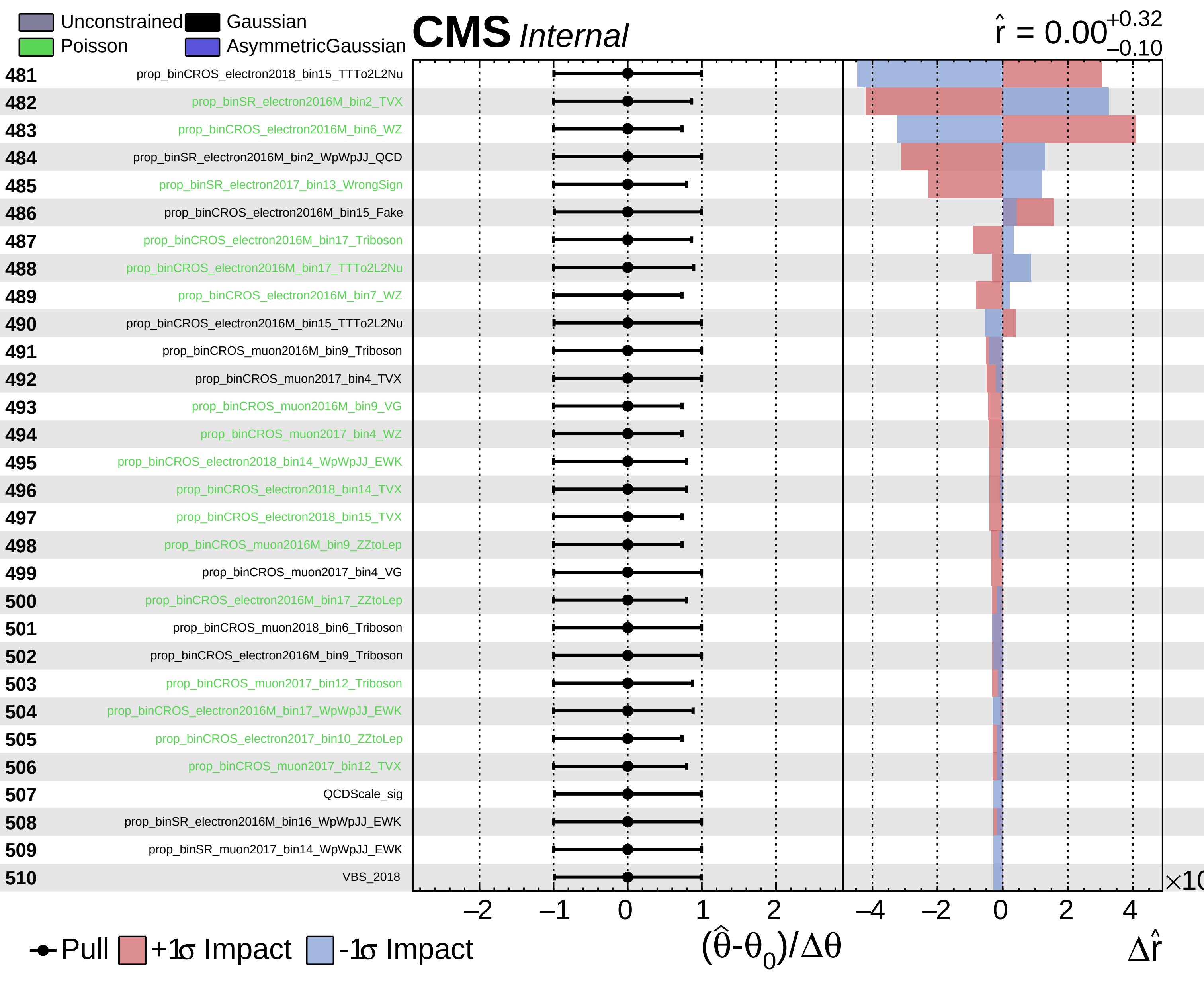
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
391 prop_binCRTT_electron2016M_bin9
392 prop_binCROS_electron2016M_bin17_WrongSign
393 prop_binCROS_electron2016M_bin10
394 prop_binCROS_muon2017_bin14
395 prop_binCROS_muon2016M_bin7
396 prop_binSR_electron2016M_bin16_WrongSign
397 prop_binCRTT_electron2016M_bin15
398 prop_binSR_electron2017_bin2
399 prop_binCROS_muon2018_bin15
400 prop_binSR_electron2016M_bin16_Triboson
401 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
402 prop_binCROS_muon2017_bin12_Fake
403 prop_binCROS_muon2016M_bin16
404 prop_binCROS_muon2017_bin15
405 prop_binSR_electron2017_bin17_WpWpJJ_QCD
406 QCDScale_ZZtoLep
407 prop_binCROS_electron2017_bin16
408 prop_binCROS_muon2016M_bin12
409 prop_binSR_electron2016M_bin17_ZZtoLep
410 lep_2018
411 prop_binCROS_muon2018_bin16
412 prop_binCROS_electron2016M_bin16
413 prop_binSR_muon2017_bin9_TTTo2L2Nu
414 prop_binCRTT_muon2016M_bin17
415 prop_binCROS_electron2018_bin13
416 prop_binCROS_electron2017_bin14
417 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
418 prop_binCROS_muon2018_bin6_TTTo2L2Nu
419 prop_binSR_electron2017_bin17_TVX
420 prop_binCROS_muon2017_bin11
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

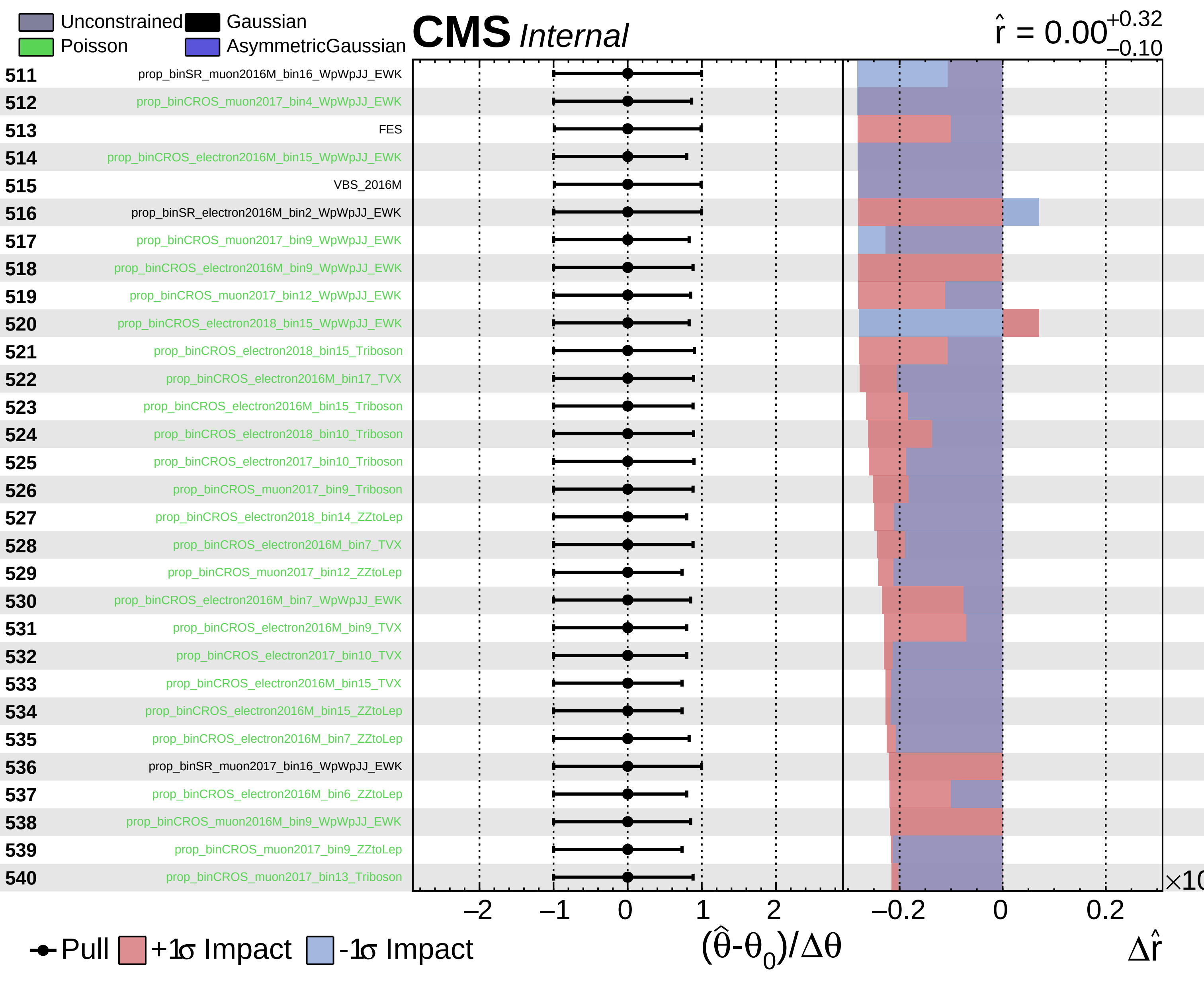


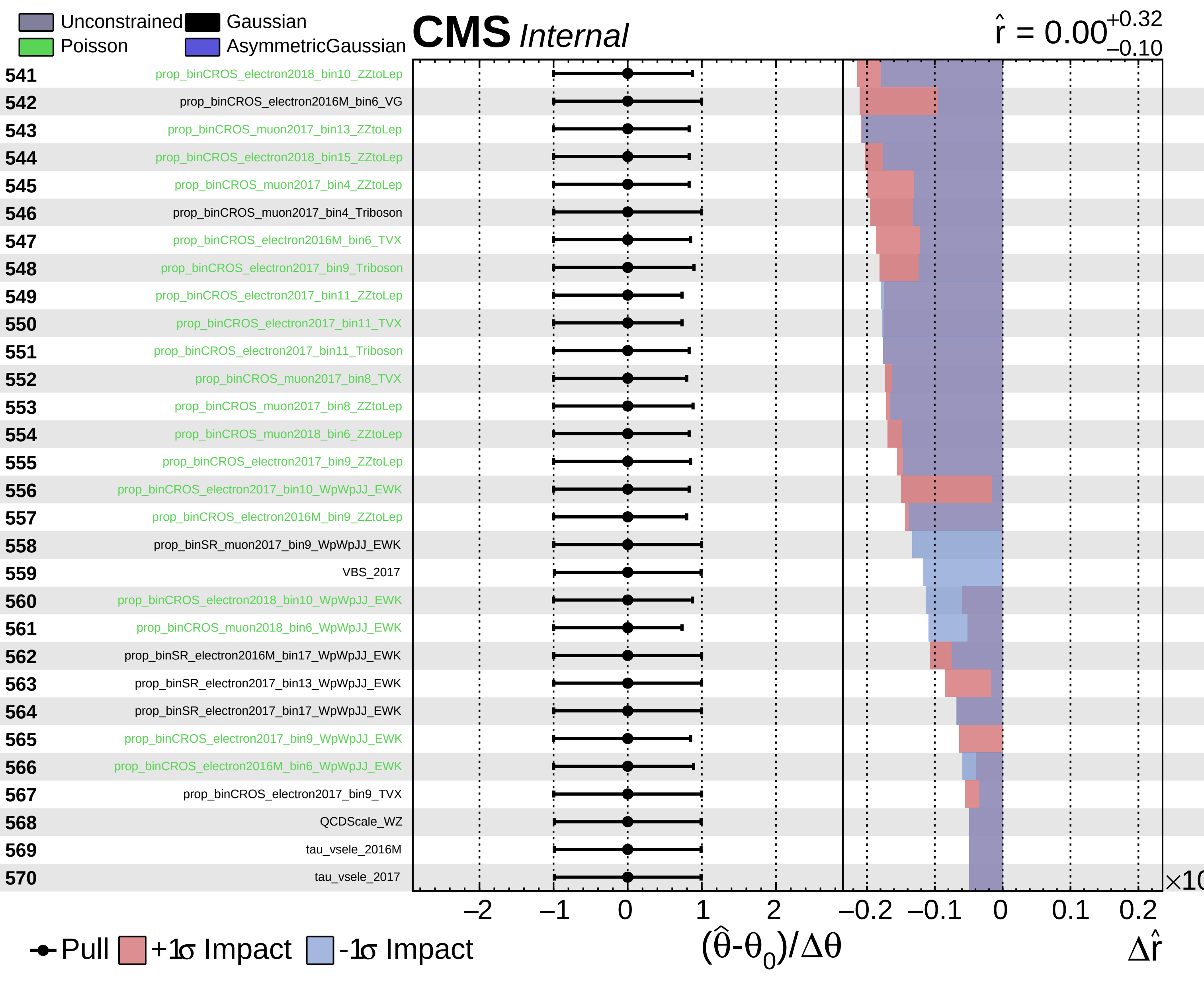
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
421 prop_binCROS_electron2018_bin15_VG
422 prop_binCRTT_electron2017_bin17
423 prop_binCROS_muon2016M_bin13
424 prop_binSR_electron2017_bin13_WpWpJJ_QCD
425 prop_binCROS_muon2017_bin8_TTTo2L2Nu
426 prop_binCROS_muon2017_bin9_TTTo2L2Nu
427 tau_vsjet_2018
428 prop_binCROS_muon2016M_bin15
429 prop_binSR_electron2016M_bin2_WZ
430 prop_binCROS_muon2017_bin13_TTTo2L2Nu
431 lep_2017
432 prop_binCROS_muon2017_bin4_TTTo2L2Nu
433 prop_binSR_electron2016M_bin17_WrongSign
434 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
435 prop_binCROS_electron2017_bin9_TTTo2L2Nu
436 prop_binSR_electron2017_bin13_Triboson
437 prop_binCROS_muon2016M_bin17
438 prop_binCRTT_electron2016M_bin17
439 prop_binCROS_muon2018_bin17
440 prop_binCROS_electron2016M_bin9_Fake
441 prop_binCROS_electron2017_bin12
442 prop_binSR_electron2016M_bin2_TTTo2L2Nu
443 prop_binCROS_electron2016M_bin13
444 prop_binCROS_muon2017_bin12_TTTo2L2Nu
445 prop_binCROS_electron2018_bin14_TTTo2L2Nu
446 prop_binSR_muon2017_bin9_WpWpJJ_QCD
447 prop_binCROS_electron2016M_bin6_Fake
448 prop_binCRTT_electron2017_bin16
449 prop_binSR_electron2017_bin13_TVX
450 prop_binSR_electron2018_bin16_WrongSign
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

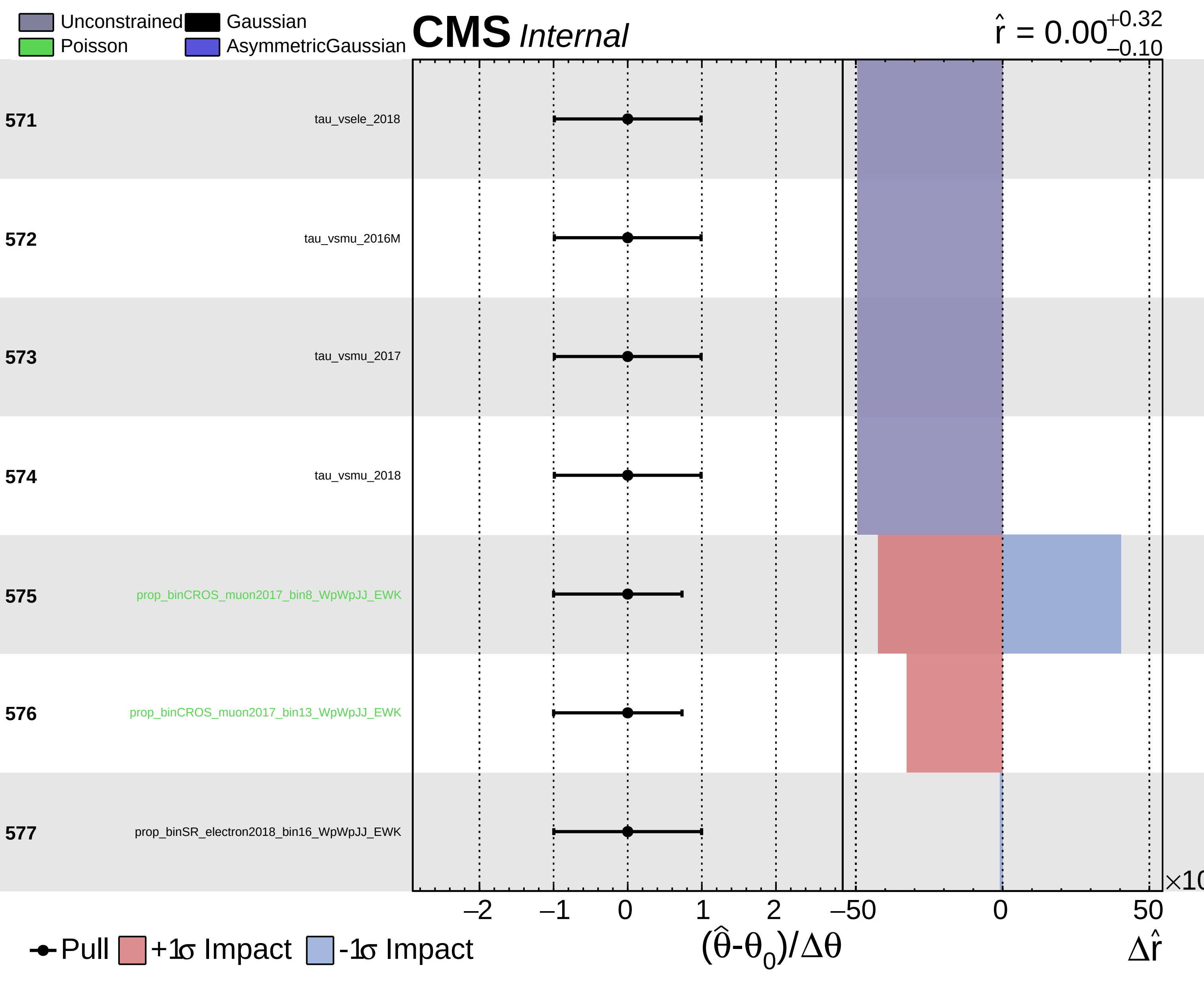


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
451 prop_binSR_muon2016M_bin16_ZZtoLep
452 prop_binSR_muon2017_bin9_TVX
453 prop_binCROS_electron2017_bin11_TTTo2L2Nu
454 lep_2016M
455 prop_binCROS_electron2017_bin17
456 prop_binCROS_electron2018_bin10_TTTo2L2Nu
457 prop_binCROS_muon2018_bin6_TVX
458 prop_binSR_electron2016M_bin16_ZZtoLep
459 prop_binCROS_electron2018_bin14_Triboson
460 prop_binSR_electron2016M_bin2_Triboson
461 prop_binCROS_electron2016M_bin7_Triboson
462 prop_binCROS_muon2017_bin9_TVX
463 prop_binCROS_muon2017_bin16
464 prop_binCROS_electron2017_bin9_WZ
465 prop_binCROS_electron2017_bin10_TTTo2L2Nu
466 prop_binCROS_electron2018_bin10_TVX
467 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
468 prop_binCROS_muon2017_bin8_Triboson
469 prop_binCROS_muon2016M_bin9_TVX
470 prop_binCROS_electron2018_bin15_Fake
471 prop_binCROS_electron2017_bin15
472 prop_binSR_electron2017_bin17_ZZtoLep
473 prop_binCROS_electron2016M_bin6_Triboson
474 prop_binSR_electron2016M_bin2_ZZtoLep
475 prop_binCROS_muon2017_bin13_TVX
476 prop_binSR_muon2017_bin9_Triboson
477 prop_binCROS_muon2017_bin8_VG
478 prop_binCROS_muon2017_bin9_VG
479 prop_binCROS_electron2017_bin11_VG
480 prop_binCROS_muon2016M_bin14
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
481 prop_binCROS_electron2018_bin15_TTTo2L2Nu
482 prop_binSR_electron2016M_bin2_TVX
483 prop_binCROS_electron2016M_bin6_WZ
484 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
485 prop_binSR_electron2017_bin13_WrongSign
486 prop_binCROS_electron2016M_bin15_Fake
487 prop_binCROS_electron2016M_bin17_Triboson
488 prop_binCROS_electron2016M_bin17_TTTo2L2Nu
489 prop_binCROS_electron2016M_bin7_WZ
490 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
491 prop_binCROS_muon2016M_bin9_Triboson
492 prop_binCROS_muon2017_bin4_TVX
493 prop_binCROS_muon2016M_bin9_VG
494 prop_binCROS_muon2017_bin4_WZ
495 prop_binCROS_electron2018_bin14_WpWpJJ_EWK
496 prop_binCROS_electron2018_bin14_TVX
497 prop_binCROS_electron2018_bin15_TVX
498 prop_binCROS_muon2016M_bin9_ZZtoLep
499 prop_binCROS_muon2017_bin4_VG
500 prop_binCROS_electron2016M_bin17_ZZtoLep
501 prop_binCROS_muon2018_bin6_Triboson
502 prop_binCROS_electron2016M_bin9_Triboson
503 prop_binCROS_muon2017_bin12_Triboson
504 prop_binCROS_electron2016M_bin17_WpWpJJ_EWK
505 prop_binCROS_electron2017_bin10_ZZtoLep
506 prop_binCROS_muon2017_bin12_TVX
507 QCDScale_sig
508 prop_binSR_electron2016M_bin16_WpWpJJ_EWK
509 prop_binSR_muon2017_bin14_WpWpJJ_EWK
510 VBS_2018
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.32 −0.10
511 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
512 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
513 FES
514 prop_binCROS_electron2016M_bin15_WpWpJJ_EWK
515 VBS_2016M
516 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
517 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
518 prop_binCROS_electron2016M_bin9_WpWpJJ_EWK
519 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
520 prop_binCROS_electron2018_bin15_WpWpJJ_EWK
521 prop_binCROS_electron2018_bin15_Triboson
522 prop_binCROS_electron2016M_bin17_TVX
523 prop_binCROS_electron2016M_bin15_Triboson
524 prop_binCROS_electron2018_bin10_Triboson
525 prop_binCROS_electron2017_bin10_Triboson
526 prop_binCROS_muon2017_bin9_Triboson
527 prop_binCROS_electron2018_bin14_ZZtoLep
528 prop_binCROS_electron2016M_bin7_TVX
529 prop_binCROS_muon2017_bin12_ZZtoLep
530 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
531 prop_binCROS_electron2016M_bin9_TVX
532 prop_binCROS_electron2017_bin10_TVX
533 prop_binCROS_electron2016M_bin15_TVX
534 prop_binCROS_electron2016M_bin15_ZZtoLep
535 prop_binCROS_electron2016M_bin7_ZZtoLep
536 prop_binSR_muon2017_bin16_WpWpJJ_EWK
537 prop_binCROS_electron2016M_bin6_ZZtoLep
538 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
539 prop_binCROS_muon2017_bin9_ZZtoLep
540 prop_binCROS_muon2017_bin13_Triboson
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
541 prop_binCROS_electron2018_bin10_ZZtoLep
542 prop_binCROS_electron2016M_bin6_VG
543 prop_binCROS_muon2017_bin13_ZZtoLep
544 prop_binCROS_electron2018_bin15_ZZtoLep
545 prop_binCROS_muon2017_bin4_ZZtoLep
546 prop_binCROS_muon2017_bin4_Triboson
547 prop_binCROS_electron2016M_bin6_TVX
548 prop_binCROS_electron2017_bin9_Triboson
549 prop_binCROS_electron2017_bin11_ZZtoLep
550 prop_binCROS_electron2017_bin11_TVX
551 prop_binCROS_electron2017_bin11_Triboson
552 prop_binCROS_muon2017_bin8_TVX
553 prop_binCROS_muon2017_bin8_ZZtoLep
554 prop_binCROS_muon2018_bin6_ZZtoLep
555 prop_binCROS_electron2017_bin9_ZZtoLep
556 prop_binCROS_electron2017_bin10_WpWpJJ_EWK
557 prop_binCROS_electron2016M_bin9_ZZtoLep
558 prop_binSR_muon2017_bin9_WpWpJJ_EWK
559 VBS_2017
560 prop_binCROS_electron2018_bin10_WpWpJJ_EWK
561 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
562 prop_binSR_electron2016M_bin17_WpWpJJ_EWK
563 prop_binSR_electron2017_bin13_WpWpJJ_EWK
564 prop_binSR_electron2017_bin17_WpWpJJ_EWK
565 prop_binCROS_electron2017_bin9_WpWpJJ_EWK
566 prop_binCROS_electron2016M_bin6_WpWpJJ_EWK
567 prop_binCROS_electron2017_bin9_TVX
568 QCDScale_WZ
569 tau_vsele_2016M
570 tau_vsele_2017
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.32}_{-0.10}$
571 tau_vsele_2018
572 tau_vsmu_2016M
573 tau_vsmu_2017
574 tau_vsmu_2018
575 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
576 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
577 prop_binSR_electron2018_bin16_WpWpJJ_EWK
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$